



FUEL DISPENSER GUIDE BOOK



PRODUCTS PASSED ISO9001 INTERNATIONAL QUALITY
MANAGEMENT SYSTEM CERTIFICATION

ATTENTION

1. Please read the guide book carefully before installation and usage.
2. Don't smoke and use the mobile phone in the gas station.
3. Don't solder and use electric tool which may cause spark in the gas station.
4. Before fueling, the engine of vehicles must be stopped.
5. The gas station must bury reliable ground wire according to the standard and the grounding resistance should be $\leq 4\Omega$.
6. Repairing and installing must be under power cut condition.
7. Before turning on the switch, please affirm the voltage of power supply system is normal.
8. To avoid overflowing, the operator is not allowed to leave when fueling.
9. Don't fuel into non-conductive polythene container and make sure to avoid causing fire for static when fueling.
10. Unprofessional personnel is forbidden to take part the machine.
11. Install the machine in well-ventilated place. Avoid the machine to be drenched and directly insolated.
12. Stop fueling and cutting the power immediately when any accident occurs suddenly.

TABLE

1. Brief Introduction	1-2
2. Illustration of Product Model	3
3. Principle of Work	3-4
4. Technical Parameters	5
5. The Structure of the Machine	6
6. Operation	
(1) The guide of operating key board	7
(2) Operation	
1) Fueling operation	8
2) Operation of setting parameter	
A. To set unit price	9
B. To set screen mode and stop mode by integer.....	10
C.To set delay of pulser	11
D.To adjust the flow rate of solenoid valve shut-off in advance.....	12
E.To configure how to start motor	13
F. To stop motor.....	14
G.To set rounding, IP and status of fuel dispenser for POS.....	15
H. To set decimal point and printer port.....	16
I.To set display port and totalizer detect.....	17
3) Inquiry the data	
A. To inquiry the total of volume and sale.....	18
B. To inquiry the shift total of volume and sale.....	18

C. To clear the shift total of volume and sale	19
D. Inquiry 2000pcs filling records	20
E. Inquiry and set time.....	21
4) Advanced options	
A. To lock liter.....	22
B. To lock amount.....	23
(3) Error code table	24
(4) Lock code table	24
7. Failure and Resolution	25-26

1. Brief Introduction

ZC series fuel dispenser is developed with modern design and latest science technology. Taking the characteristics of perfect performance, reliable quality and beauty figure, ZC series fuel dispenser is a kind of advanced meter equipment for vehicles.

During production, the machine strictly complys with 《GB/T9081-2008 Fuel dispensers for motor vehicles 》 , 《 JJG443-2015 Verification regulation of fuel dispensers 》 and 《GB22380.1-2012 Explosion protected safety technique of the petrol filling station-Part 1: Explosion protected safety technique requirements for fuel filling dispenser》 .

The machine takes advantage of high accuracy, convenient operation and easy repair, and it also bears abroad functions, such as fueling as preset data, checking ata and accumulative amount of fuel and sum of sales. The data is counted by Mini-computer and displayed on the LCD screen, which can be kept in the computer for 7 years. The machine is widely used in gas station for vehicles, suitable for gasoline, diesel oil, kerosene and various light fuel.

The machine is equipped with communication ports, by which computer can connect 16pcs nozzles of fuel dispensers. By computer, we can monitor and control more than one fuel dispenser working synchronously. Connected with relative equipments, the computer can print out the fueled amount and sum of each machine, carrying out real-time management to stock oil and IC card charge, therefore realizing the gas station modern administration.

ZC series fuel dispenser adopts ex-proof structure design, has passed the test of appointed

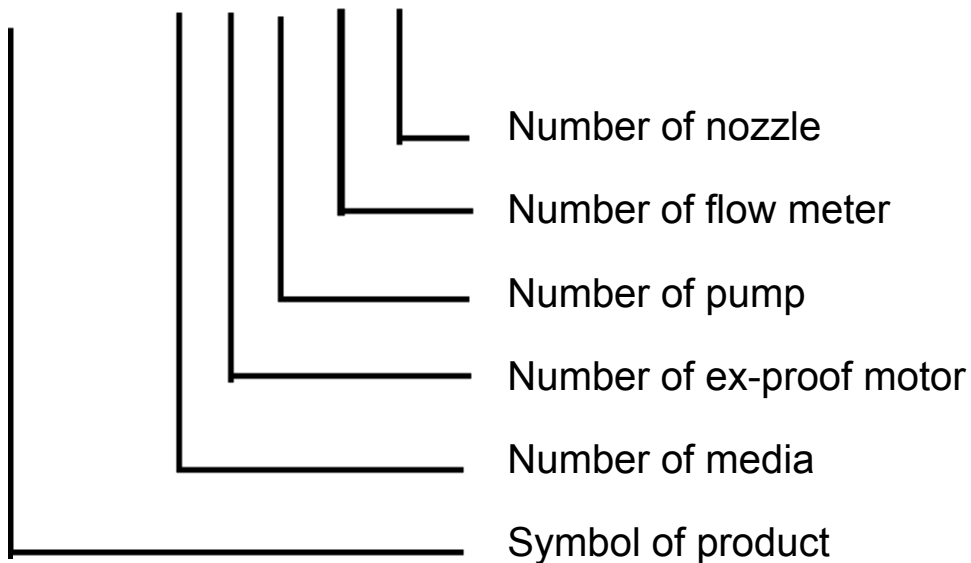
inspection department, achieved the <Ex-proof certificate>, <Meter approval certificate>, <Fuel dispenser manufacturing license>. The machine is safe for operation, which can work under bad environmental condition (extremely cold, hot and moist).

At present, ZC series has models: single nozzle, double nozzles, four nozzles and eight nozzles etc, based on which we developed luxurious type(including high hang, stainless steel side, and various big flux type, there are more than 30 types totally), so it can meet customer's different requirements.

Operation, inner structure and working principle of ZC series fuel dispenser are nearly same, so the guide book mainly introduces model of single nozzle for reference. If any difference, we have detailed explanations and diagrams in the guide book.

2. Illustration of Product Model

ZC - □ □ □ □ □



3. Principle of Work

Firstly, motor starts and drives the pump working, then the pump sucks the oil from tank into the pump cavity, where the oil is pressured into the oil-gas separator which is combined in pump, at the same time air is separated from the oil and vented out. Secondly the separated oil without air comes into the flow meter which driving the metric piston moving, and then angle displacement signals corresponding to the accurately-metered oil volume is transmitted, subsequently the pulser converts angle displacement signals into pulse signals which are sent to the mini-computer controller. That records and displays the present-time oil amount(sum) or accumulative amount(sum). During this course, the metered oil is transported into the vehicles through electromagnetic valve, observation apparatus and nozzle.

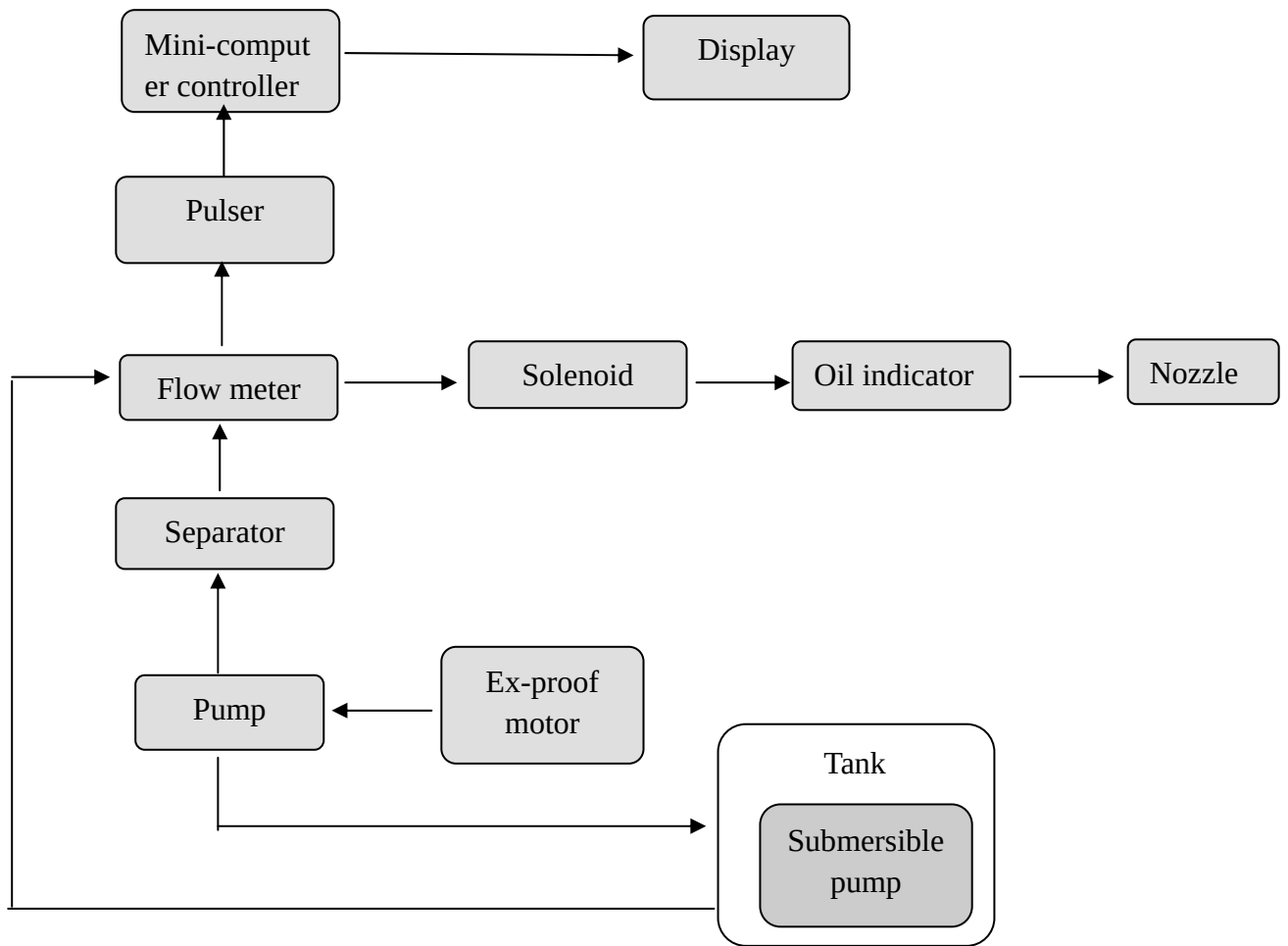


Diagram 1

Principle Diagram of ZC series Fuel Dispenser

4. Technical Parameters

- (1) Temperature: $-20^{\circ}\text{C}\sim+55^{\circ}\text{C}$
- (2) Humidity: 30%~90%
- (3) Air pressure: 86~106 kPa
- (4) Max. flow rate: 5~50L/min or 5~90L/min(optional)
- (5) Calculation accuracy: $\pm 0.30\%$
- (6) Repeating tolerance: $\leq 0.15\%$
- (7) Inlet pressure: $\geq 54\text{kPa}$
- (8) Outlet pressure: $\leq 0.30\text{MPa}$
- (9) Suction Distance: 6m(Vertical), 50m(Horizontal)
- (10) Voltage: 110V/220V/380V(50/60Hz)
- (11) Power: 1HP(0.75kW)
- (12) Noise: $\leq 80\text{dB(A class)}$
- (13) Unit price range: 0.1~999.99
- (14) Single volume range: 0.01~999999.99
- (15) Accumulative volume range: 0.01~9999999999.99
- (16) Working life: $\geq 1.5 \times 10^6 \text{ L}$
- (17) Oil pipe: Inner diameter: $\phi 19\text{mm}$, outer diameter: $\phi 30\text{mm}$
- (18) Ex-proof signal: Ex d mb II A T3 Gb

5. The Structure of the Machine

(1) Electronic controller

(2) Solenoid valve

(3) Flow meter

(4) Oil indicator

(5) Pump

(6) Nozzle booter

(7) Oil pipe

(8) Flexible pipe

(9) Plated body

(10) Keyboard

(11) Pulser

(12) Ex-proof junction box

(13) Ex-proof motor

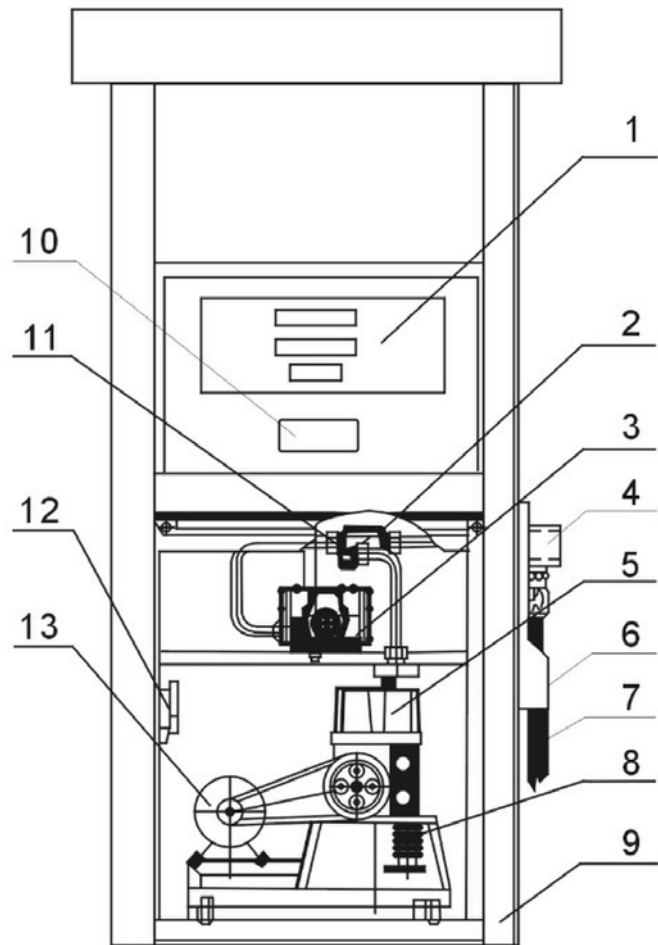


Diagram 2

6. Operation

(1) The guide of operating key board

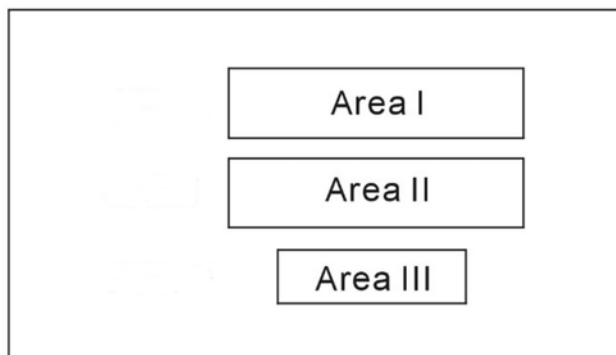


Diagram 3

The layout diagram of display windows

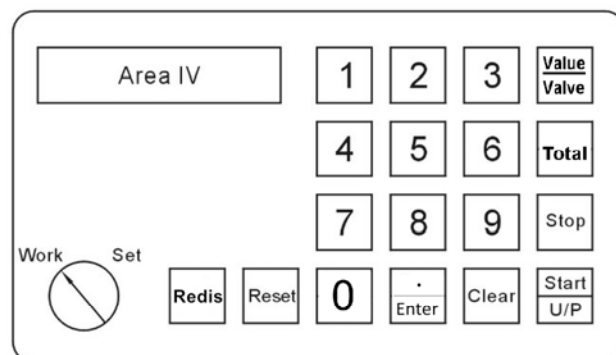


Diagram 4

The layout diagram of keyboard for ZC series

Use Keyboard	State		
		Work	Set
Start/U/P		Start fueling	Set unit price
Stop		Stop machine	Stop machine
<u>Value</u> Valve		Switch between “L” and “P”	Set value or solenoid valve
Total		Check accumulative amount of fuel and sum of sales each class	Check accumulative amount of fuel and sum of sales
●/Enter		Radix point / stop by integer	Confirmation
Redis		Display present-time amount of injected fuel and sum of sales when power cuts suddenly. (Redis=redisplay)	Combination button or amend the time or check filling date
The position of key		Ready to fuel	Set parameters
Keypad: 0~9		Value number (Litre stops by integer when pressing “0” under working)	
Clear		Clear data	
Reset		Restart machine when errors occur	

(2) Operation

1) Fueling operation

Operation	Method
Fueling randomly	Turn the key to "Work"  - Press  or take up nozzle under Area IV displaying "L 0" or "p 0". - Open the nozzle to start fueling, Area I,II read from "0". - Press "Stop" or close & hang up the nozzle to stop fueling, Area I,II display present-time amount of injected fuel and sum of sales.
Fueling as preset data of "Litre"	Turn the key to "Work"  - Area IV display "L". If "P", press  to switch to "L". - Input the present data, Area IV displays the same preset data at the same time. - Press  or take up the nozzle, Area I,II display "0", Area IV remains unchangeable. - Open the nozzle to start fueling, Area I,II read real-time data and stop fueling when reaching the preset value. - Hang up the nozzle, Area I,II display present-time amount of injected fuel and sum of sales.
Fueling as preset data of "Sale"	Area IV should be switched to "P", the other operations are same as above.

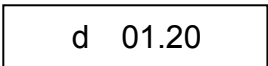
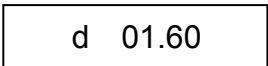
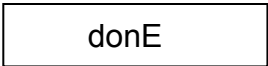
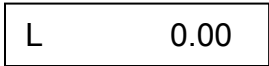
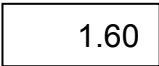
Remark

- If the preset data of two times are same, we have to input the same preset value again.
- Stop the machine before reaching the preset value, it displays the present-time amount of injected fuel and sum of sales.

2) Operation of setting parameter

A. To set unit price

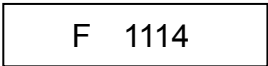
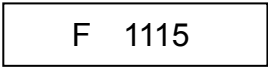
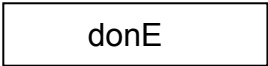
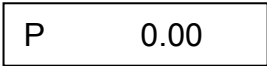
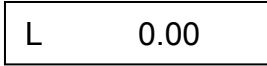
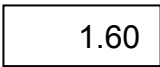
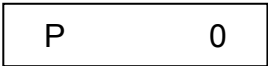
Setting range 0.01~999.99. For example, to input a new price 1.60.

Operation	Display of Area IV	Display of Area I, II, III
1) Turn the key to “set”	Remain unchangeable	Remain unchangeable
2) Press 	 Display the present-time unit price	Remain unchangeable
3) Press  firstly, and input new value “160”	 Display as above	Remain unchangeable
4) Press  to confirm	 Display as above	  Display new unit price 
5) Turn the key to “Work” and wait for fueling	 Display as above	Remain unchangeable

Remark

1. When inputting new unit price, we should input one integer and two decimals, but do not need to input radix point.

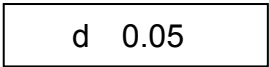
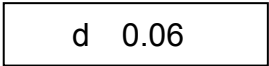
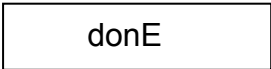
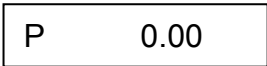
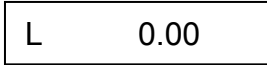
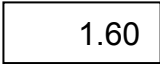
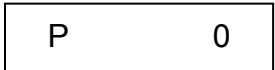
B. To set screen mode and stop mode by integer.

Operation	Display of Area IV	Display of Area I, II, III
1) Turn the key to "set"	Remain unchangeable	Remain unchangeable
2) Long press  and press  together Default: Y 1114	 Display as above	Remain unchangeable
3) Press  firstly, and then input new value "1115"	 Display as above	Remain unchangeable
4) Press 	 Display as above	  
5) Turn the key to "Work" and wait for fueling	 Display as above	Remain unchangeable

Remark

1st digit: 0=664,1=885
2nd digit: 1=Amount/Liter, 0= Liter/Amount
3rd digit: Thai=0, English=1
4th digit: Stop integer
2=at Unit
3=at 5
4=at 10
5=at 50
6=at 100
7=at 500
8=at 1000
9=at 10000

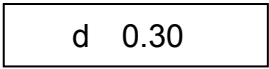
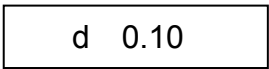
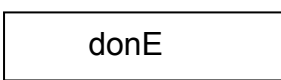
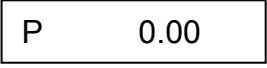
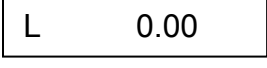
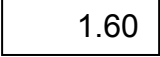
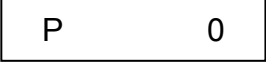
C. To set delay of pulser

Operation	Display of Area IV	Display of Area I, II, III
1) Turn the key to "set"	Remain unchangeable	Remain unchangeable
2) Long press  and press  together Default: d 0.05	 Display as above	Remain unchangeable
3) Press  firstly, and then input new value "6"	 Display as above	Remain unchangeable
4) Press 		  
5) Turn the key to "Work" and wait for fueling	 Display as above	Remain unchangeable

Remark

1. Value Range: 0.05~0.10
2. Filling data less than default value will be shown on LCD when machine stops and added to total

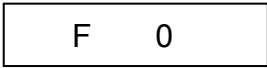
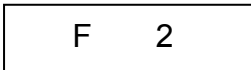
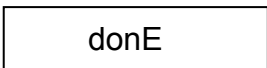
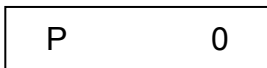
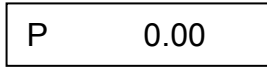
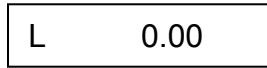
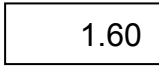
D. To adjust the flow rate of solenoid valve shut-off in advance.

Operation	Display of Area IV	Display of Area I, II, III
1) Turn the key to "set"	Remain unchangeable	Remain unchangeable
2) Press  Default: 0.30	 Display as above	Remain unchangeable
3) Press  firstly, and then input new value "10"	 Display as above	Remain unchangeable
4) Press 	 Display as above	  
5) Turn the key to "Work" and wait for fueling	 Display as above	Remain unchangeable

Remark

New value range :0.01-0.99

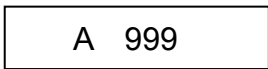
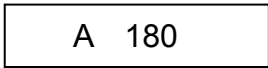
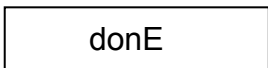
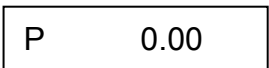
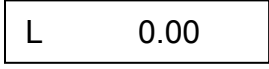
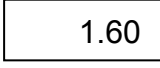
E. To configure how to start motor

Operation	Display of Area IV	Display of Area I, II, III
1) Turn the key to “set”	Remain unchangeable	Remain unchangeable
2) Long press  and press  together Default: d 0	 Display as above	Remain unchangeable
3) Press  firstly, and then input new value “2”	 Display as above	Remain unchangeable
3) Press 	 Display as above	Remain unchangeable
4) Turn the key to “Work” and wait for fueling	 Display as above	  

Remark

0=by start button or lift up nozzle
1=Start button only
2=Lift up nozzle only

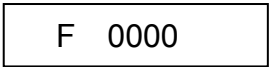
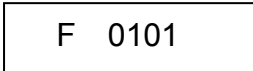
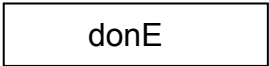
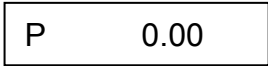
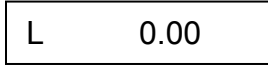
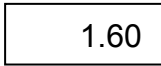
F. To stop motor

Operation	Display of Area IV	Display of Area I, II, III
1) Turn the key to “set”	Remain unchangeable	Remain unchangeable
2) Long press  and press  together Do the same operation again as above to enter condition 2	 Display as above	Remain unchangeable
3) Press  firstly, and then input new value “180”	 Display as above	Remain unchangeable
4) Press 	 Display as above	  
5) Turn the key to “Work” and wait for fueling	 Display as above	Remain unchangeable

Remark

- Condition 1 (A 999):
motor stops when start or nozzle up, there is no oil within max 999 seconds.
- Condition 2 (b 999):
motor stops after nozzle is closed, there is no oil within max 999 seconds.

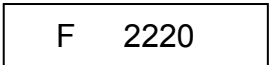
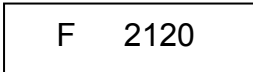
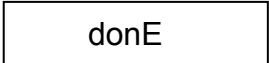
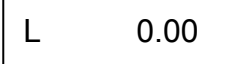
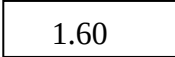
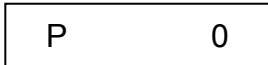
G. To set rounding, IP and status of fuel dispenser for POS.

Operation	Display of Area IV	Display of Area I, II, III
1) Turn the key to "set"	Remain unchangeable	Remain unchangeable
2) Long press  and press  together Default: Y 0000	 Display as above	Remain unchangeable
3) Press  firstly, and then input new value "0101",	 Display as above	Remain unchangeable
4) Press 	 Display as above	  
5) Turn the key to "Work" and wait for fueling	 Display as above	Remain unchangeable

Remark

1ST digit: 0=rounding, 1-9=without rounding
2nd digit: 0=offline, 1=manage, 2=monitor, 3=all
3rd 4th digit: address

H. To set decimal point and printer port.

Operation	Display of Area IV	Display of Area I, II, III
1) Turn the key to "set"	Remain unchangeable	Remain unchangeable
2) Long press  and press  together Default: d 2220	 Display as above	Remain unchangeable
3) Press  firstly, and then input new value "2120",	 Display as above	Remain unchangeable
3) Press  Enter	 Display as above	  
4) Turn the key to "Work" and wait for fueling	 Display as above	Remain unchangeable

Refer to the following table to select code for required display format:

code	Litre	Sale	Price
200	2	0	0
210	2	1	0
201	2	0	1
211	2	1	1
221	2	2	1
212	2	1	2
222	2	2	2
232	2	3	2
223	2	2	3
233	2	3	3

code	Litre	Sale	Price
300	3	0	0
310	3	1	0
301	3	0	1
311	3	1	1
321	3	2	1
312	3	1	2
322	3	2	2
332	3	3	2
323	3	2	3
333	3	3	3

Remark

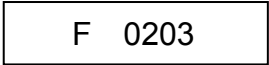
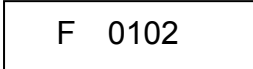
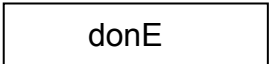
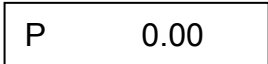
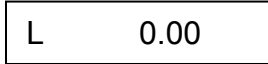
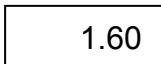
4th digit: printer

0=printer 1(nozzle 1), printer 2 (nozzle 2)

1=nozzle 1&2 share printer 1

2=nozzle 1&2 share printer 2

I. To set display port and totalizer detect.

Operation	Display of Area IV	Display of Area I, II, III
1) Turn the key to "set"	Remain unchangeable	Remain unchangeable
2) Long press  and press  together Default: d 0203	 Display as above	Remain unchangeable
3) Press  firstly, and then input new value "0102"	 Display as above	Remain unchangeable
4) Press 	 Display as above	  
5) Turn the key to "Work" and wait for fueling	 Display as above	Remain unchangeable

Remark

1st digit: U/P(0=non-sync, 1= sync)

2nd digit: bard rate(0=4800, 1=9600, 2=19200)

3rd digit: display

0=display1(keypad 1), display 2 (keypad 2)

1=display1, display 2 (keypad 1)

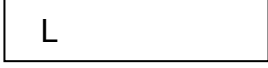
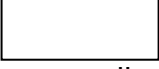
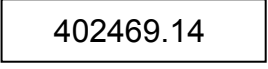
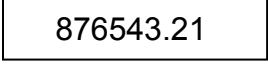
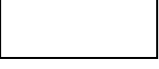
2=display1, display 2 (keypad 2)

Model 1&2 is used for only single nozzle

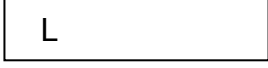
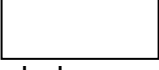
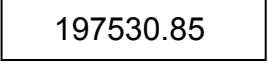
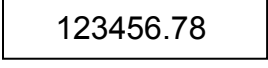
4th digit: totalizer (0=detect, 1=nozzle 1 detect off, 2=nozzle 2 detect off, 3=non-detect)

(3) Inquiry the data

A. To inquiry the total of volume and sale.

Operation	Display of Area IV	Display of Area I, II, III
1) Turn the key to “set”	Remain unchangeable	Remain unchangeable
2) Press 	 Display as above	Display 3 seconds firstly    And then display secondly    Sale: 1402469.14 Litre: 876543.21

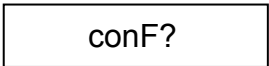
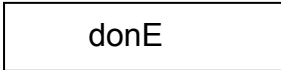
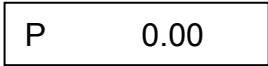
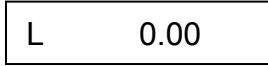
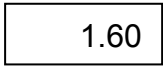
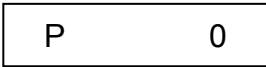
B. To inquiry the shift total of volume and sale

Operation	Display of Area IV	Display of Area I, II, III
1) Turn the key to “work”	Remain unchangeable	Remain unchangeable
2) Press 	Remain unchangeable	Display 3 seconds firstly    And then display below   

Remark

1. If the data is beyond 8 digits, it displays front four digits firstly and then display last 8 digits accordingly.

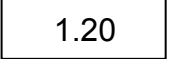
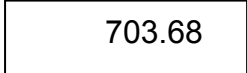
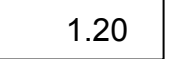
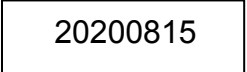
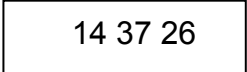
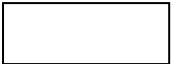
C. To clear the shift total of volume and sale.

Operation	Display of Area IV	Display of Area I, II, III
1) Turn the key to "set"	Remain unchangeable	Remain unchangeable
2) Long press  and press  together	 Display as above	Remain unchangeable
3) Input password 12345678 And press 	 Display as above	Remain unchangeable
4) Press 	 Display as above	  
5) Turn the key to "Work" and wait for fueling	 Display as above	Remain unchangeable

Remark

1. If password is not correct, area IV will show (Err) for 3 seconds. It will show (n 0), then input password again.

D. Inquiry 2000pcs filling records.

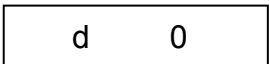
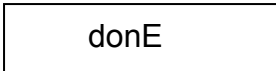
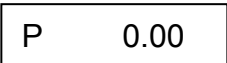
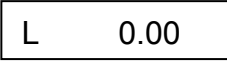
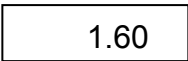
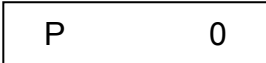
Operation	Display of Area IV	Display of Area I, II, III
1) Turn the key to "Work"	Remain unchangeable	Remain unchangeable
2) Long press  and press  together	 Display as above	   Display the present-time value
3) Press  firstly and input new value "10", then press 	 Display as above	   Display the tenth filling record
4) Press  to show filling date.	Remain unchangeable	   YYYY/MM/DD HH/MM/SS
5) Press  to printer the inquired filling record.	Remain unchangeable	Remain unchangeable

Remark

Fast operation(for previous 10th filling records):

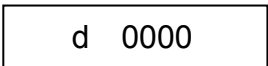
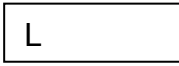
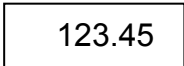
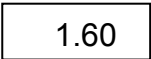
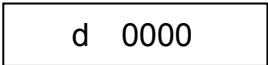
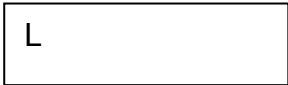
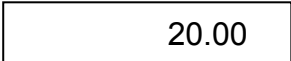
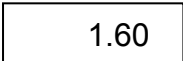
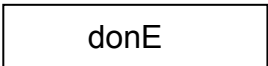
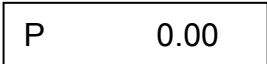
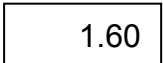
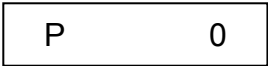
1. Turn key to "work"
2. Long press "Redis" + "x" (x= 0-9)

E. Inquiry and set time.

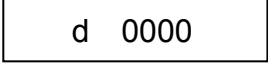
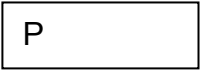
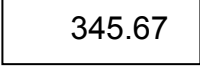
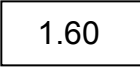
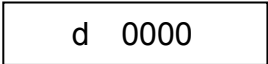
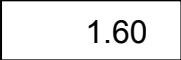
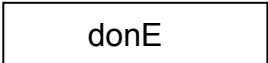
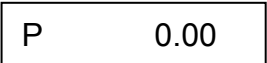
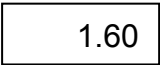
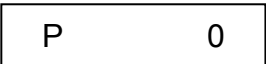
Operation	Display of Area IV	Display of Area I, II, III
1) Turn the key to “set”	Remain unchangeable	Remain unchangeable
2) Press  for 3 seconds	 Display as above	Show real-time dynamic time
3) Press 	 Display as above	Show current static time
4) Input new value, then press  format: YY/MM/DD/HH/MM/	 Display as above	  
5) Turn the key to “Work” and wait for fueling	 Display as above	Remain unchangeable

(4) Advanced options

A. To lock liter

Operation	Display of Area IV	Display of Area I, II, III
1) Turn the key to “set”	Remain unchangeable	Remain unchangeable
2) Long press  and press  together	 Display as above	 Show total current litres, for example “12345”  
3) Input new value “20”,	 Display as above	  
4) Press 	 Display as above	  
5) Turn the key to “Work” and wait for fueling	 Display as above	Remain unchangeable

B. To lock amount

Operation	Display of Area IV	Display of Area I, II, III
1) Turn the key to "set"	Remain unchangeable	Remain unchangeable
2) Long press  and press  together	 Display as above	 Show total current amount for example "34567"  
3) Input new value "30"	 Display as above	  
4) Press 	 Display as above	  
5) Turn the key to "Work" and wait for fueling	 Display as above	Remain unchangeable

(3) Error code table

Error code	Information	Solution
E01	No pulser or pulser error	1. Examine pulser & cable 2. replace pulser
E02	There is no oil discharge over fixed range when beginning to fuel	hang up nozzle
E03	There is no oil discharge over fixed range when nozzle is closed	hang up nozzle
E04	None	None
E05	Total liter is full	Clear data
E06	Total Amount is full	Clear data
E07	Totalizer disable	1. examine totalizer & cable 2. replace totalizer
E12	Range is beyond	Input new value
E13	Busy	Set parameters when another nozzle stop working

(4) Lock code table

Lock code	Information	Solution
Lock 3	Lock liter	Unlock
Lock 4	Lock amount	Unlock
Lock 5	None	None
Lock 6	Set IP address, but failed to send 2000pcs filling records to POS system	1. clear IP address 2. Connect POS and send data
Lock M	Start motor mode	Check the mode
Lock 8	Calibration lock by dip switch K1	Operation after dip switch on
Lock 0	Initial lock by dip switch K2	Operation after dip switch on

7.Failures and solutions

Failure	Reason	Solution
No oil when power on	1. electrical system fails to open electromagnetic valve 2. belt of motor looses 3. pump or flow meter fails, or filter is blocked 4. pump or low-pressure cavity of oil-gas separator is mixed with air 5. inputting oil pipeline system leaks air 6. motor cycle backward 7. electromagnetic valve fails	1. examine and repair electrical system 2. adjust belt of motor 3. examine and repair pump, flow meter or clean filter 4. examine and repair relevant airproof mat or big floater valve 5. examine and repair pipeline system 6. alter the connection of two wire of motor randomly 7. replace electromagnetic valve
Output of oil reduces greatly	1. same as above-mentioned 2,5 2. 2.filter is blocked 3. inputting oil pipeline system is blocked or pipe diameter doesn't comply with standard 4. nozzle is blocked	1. same as above-mentioned 2,5 2. clean filter 3. clean or replace pipe 4. clean nozzle
Metering is inaccurate	1. joint board of flow meter wars to increase the distance of piston 2. inner parts of flow meter wears seriously	1. replace the joint board 2. replace flow meter
Vent-pipe leak oil	1. returning-oil floater is worn and works abnormally 2. pipeline is mounted too longly 3. tank is built too highly	1. examine and repair returning-oil floater 2. replace pipeline 3. block up vent-hoe
Oil is output with air bubble	1. low-pressure cavity of flow meter is mixed with air 2. damping slot or vent-hole of flow meter is blocked 3. inputting oil pipe leaks air	1. examine and repair relevant airproof mat or flange 2. examine and repair outputting air valve 3. examine and repair inputting oil pipe
Nozzle leaks oil	1. mandril of nozzle leaks oil 2. spout of nozzle leaks oil	1. screw down back nut or replace air proof loop 2. screw down nut of nozzle spout
No display when power on	1. plug-in between display board and mainboard doesn't connect well 2. power board cannot output 5V	1. connect the plug-in tightly 2. examine and repair power board
Redisplay is not light enough or invalidated	1. battery without power 2. the key of redisplay fails	1. replace battery 2. replace the key of redisplay

	3. mainboard fails	3. replace mainboard
Pump start up, and nozzle doesn't fuel, but the display counts	1. outputting oil pipeline of flow meter leaks oil 2. electromagnetic valve leaks air	1. check outputting oil pipeline of flow meter 2. check electromagnetic valve
Motor can't work as normal	1. power supply is not normal 2. wire of motor is connected wrong 3. motor fails 4. relay fails	1. check whether the power supply is normal 2. screw down connection pole 3. replace motor 4. replace relay
Fuel without counting	1. pulser fails 2. mainboard fails	1. replace pulser 2. replace mainboard
Without display or with continuous data when pressing keys	Key-presss or keyboard fails	Replace key-press or keyboard
Fail to record unit price, accumulative liter and so on	Battery fails	Replace battery
No stop when reaching the ration oil or amount	1. replay fails 2. mainboard fails	1. replace replay 2. replace mainboard
The date on display doesn't comply with actual fueled oil	1. axes of pulser looses 2. coupler of photoelectricity fails	1. repair axes 2. replace coupler of photoelectricity

Eliminate the fault:

